2 **(a)** the (mutual) (gravitational) force of attraction between two <u>point masses</u> is proportional to the product of their masses & inversely proportional to the square of their separation. [1]

(b) 'Gravity' refers to the gravitational <u>force</u> of attraction acting on an object [1]

while gravitational field strength is gravitational <u>force per unit mass</u> at that point (towards the centre of Earth). [1]

(c) $g = \frac{GM_E}{r^2}$, hence $g \alpha \frac{1}{r^2}$ [1]

$\frac{g'}{g} = \left(\frac{6.38 \times 10^6}{6.38 \times 10^6 + 0.12 \times 10^6}\right)^2$ [1]

$g' = 9.45 \text{ N kg}^{-1}$ [1]

(d) (The astronaut and spacecraft have the same angular velocity and orbital radius.) (Therefore) they have the same <u>centripetal acceleration</u> / <u>acceleration towards (centre of) Earth</u> / <u>acceleration of free fall</u> (since $a = r\omega^2$). [1]

(Centripetal force is only provided by gravitational force.) Hence contact force is zero and the astronaut feels weightless. [1]

(e) Apparent weight, $N = F_G - ma_c$, where F_G is the gravitational force on the person by the Earth and a_c is the centripetal acceleration. [1]

There is no centripetal acceleration at the pole but there is at the equator; hence, the apparent weight is higher at the pole (while that at the equator is smaller by the centripetal force at the equator). [1]

DO NOT ACCEPT: Earth is bulging at the centre.